

Applied Math II: Statistical Learning

Final exam, 2026.03.17

Name:

You have 120 minutes to complete the exam. You may use a sheet of notes, but no computers or textbooks are allowed. Please write legibly, and indicate clearly if your solution is continued on another sheet of paper. Good luck!

1. Supervised vs. unsupervised learning

(3 points) In 1-3 sentences, explain the difference between supervised and unsupervised learning, and give an example of each that we have covered in class.

2. Normal distribution

(6 points) Let $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be a multivariate random variable. Let $\mathbf{z} = \mathbf{A}\mathbf{x} + \mathbf{b}$, where $\mathbf{A}$ is any arbitrary matrix of suitable size, and $\mathbf{b}$ is a vector. What is the probability distribution $p(\mathbf{z})$? Show your work to justify your result mathematically.

3. Regression with Tikhonov regularization

Suppose that we want to do linear regression with the following loss function:

$$E(\mathbf{w}) = \sum_n [\mathbf{w} \cdot \mathbf{x}_n - t_n]^2 + \lambda |\mathbf{\Gamma} \mathbf{w}|^2, \quad (1)$$

where $\mathbf{\Gamma} \in \mathbb{R}^{M \times M}$, and $|\cdot|^2$ is the L2 norm. This problem is known as regression with Tikhonov regularization.

(a) (4 points) Derive the expression for the weight vector $\mathbf{w}^*$ that minimizes the above loss function.

(b) (3 points) Give a simple, sufficient, and necessary condition on $\mathbf{\Gamma}$ that guarantees that $E(\mathbf{w})$ has a unique minimum $\mathbf{w}^*$. The condition should involve only $\mathbf{\Gamma}$ and should hold for arbitrary data $\{(\mathbf{x}_1, t_1), \dots, (\mathbf{x}_N, t_N)\}$.

(c) (3 points) Recall the Bayesian justification for ridge regression, where the regularization term is interpreted as a prior imposed on the parameters. What is the prior distribution for $\mathbf{w}$ that gives us the loss function $E(\mathbf{w})$?

(d) (4 points) Suppose you solve a Tikhonov regularization problem in a two-dimensional feature space ($M = 2$) and obtain a weight vector $\mathbf{w}^*$ that minimizes $E(\mathbf{w})$. The solution lies on an isocontour of $|\mathbf{X}\mathbf{w} - \mathbf{t}|^2$, where the rows of $\mathbf{X}$ and $\mathbf{t}$ correspond to data points $\mathbf{x}_n$ and t_n , respectively. It also lies on the contour of $|\mathbf{\Gamma}\mathbf{w}|^2$. Draw a diagram that plausibly depicts these two isocontours, in a case where $\mathbf{\Gamma}$ is *not* diagonal and $\mathbf{t} \neq 0$. You should label which isocontour corresponds to $|\mathbf{X}\mathbf{w} - \mathbf{t}|^2$, which corresponds to $|\mathbf{\Gamma}\mathbf{w}|^2$, and the point $\mathbf{w}^*$. The elements must appear in a plausible geometric relationship to each other.

4. Multiclass classifier gradient

(4 points) For K classes, one-hot targets t_{nk} , and activations

$$a_{nk} = \mathbf{w}_k^\top \boldsymbol{\phi}_n, \quad y_{nk} = \frac{e^{a_{nk}}}{\sum_{j=1}^K e^{a_{nj}}}, \quad (1)$$

where $\mathbf{w}_k, \boldsymbol{\phi}_k \in \mathbb{R}^M$. The cross-entropy loss function is:

$$E(\{\mathbf{w}_k\}) = - \sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk}. \quad (2)$$

Derive $\nabla_{\mathbf{w}_k} E$ and write the update equation for implementing gradient descent.

5. Distance to hyperplane

(4 points) Let the decision boundary of a linear classifier be $w^\top x + b = 0$ with $w \neq 0$. Show the signed distance from a point x to the hyperplane is $d(x) = \frac{w^\top x + b}{\|w\|}$.

6. Covariance M-step mistake

(4 points) A student writes the M-step covariance update for a Gaussian mixture model:

```
def update_covariances(X, r, mu):
    # X: (N,M), r: (N,K), mu: (K,M)
    N, M = X.shape
    K = r.shape[1]
    Sigma = np.zeros((K, M, M))
    for k in range(K):
        Nk = r[:, k].sum()
        for n in range(N):
            Sigma[k] += r[n, k] * np.outer(X[n], X[n])
        Sigma[k] /= Nk
    return Sigma
```

Identify the conceptual bug and provide the corrected update inside the loops (just the essential change).

7. HMM transition update

In EM for an HMM, recall that

$$\xi_t(i, j) = p(z_t = i, z_{t+1} = j \mid x_{1:T}) \quad (1)$$

and

$$\gamma_t(i) = p(z_t = i \mid x_{1:T}). \quad (2)$$

Recall that, in the M step, we maximize Q , i.e. the expectation of the complete data likelihood, with respect to the model parameters, where

$$Q = \sum_{k=1}^K \gamma(z_{1k}) \ln \pi_k + \sum_{n=2}^N \sum_{j,k=1}^K \xi(z_{n-1,j}, z_{nk}) \ln A_{jk} + \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \ln p(\mathbf{x}, \vec{\phi}_k). \quad (3)$$

(a) (2 points) What is the constraint that A must satisfy?

(b) (6 points) Using your answer from (a) and the above equation, derive the equation for A_{ij}^{new} , the updated transition probability matrix.